Table 2.1 The dimensionless scaled variables and parameters that are used to find the initial amount of scaled reserve.

$\tau = a\dot{k}_M$	$\tau_b = a_b\dot{k}_M$	$l = L/L_m$	$l_b = L_b/L_m$
$u_E = \frac{E}{g[E_m]L_m^3}$	$u_E^0 = \frac{E_0}{g[E_m]L_m^3}$	$u_H = \frac{E_H}{g[E_m]L_m^3}$	$u_H^b = \frac{E_H^b}{g[E_m]L_m^3}$
$e = gu_E/l^3$	$e_b = gu_E^b/l_b^3$	$e_H = gu_H/l^3$	$e_H^b = gu_H^b/l_b^3$
$x = \frac{g}{e+g}$	$x_b = \frac{g}{e_b+g}$	$\alpha = 3gx^{1/3}/l$	$\alpha_b = 3gx_b^{1/3}/l_b$
$y = \frac{xe_H}{1-\kappa}$	$y_b = \frac{x_b e_H^b}{1-\kappa} = gx_b v_H^b l_b^{-3}$	$v_H^b = \frac{u_H^b}{1-\kappa}$	$k = \dot{k}_J/\dot{k}_M$

for growth in 'demand' systems. Later, on {173}, I discuss deviations from the von Bertalanffy growth curve that can be understood in the context of the present theory.

In contrast, at low food densities, fluctuations in food density soon induce deviations from the von Bertalanffy curve. This phenomenon is discussed further in the section on genetics and parameter variation, {296}. Growth ceases, i.e. $\frac{d}{dt}V = 0$, if the reserve density equals a threshold value, $[E] = [\dot{p}_S]L/\kappa\dot{v}$.

2.6.2 States at birth and initial amount of reserve

During the embryo stage, dry weight decreases, but the amount of structure increases: reserve is converted into structure. The initial amount of reserve is not a free parameter because the reserve density, i.e. the ratio of the amounts of reserve and structure, at birth tends to covary with that of the mother at egg production; well-fed mothers give birth to well-fed offspring. Such maternal effects are typical and have been found in e.g. birds [824], reptiles, amphibians [715], fishes [478], insects [770, 994, 993], crustaceans [411], rotifers [1287], echinoderms and bivalves [98]. Maternal effects explain, for instance, why the batch fecundity of the anchovy *Engraulis* increases during the spawning season in response to a decrease in food availability [877]. However, some species seem to produce large eggs under poor feeding conditions, e.g. some poeciliid fishes [955], daphnids [413] and *Sancassania* mites [82]. Moreover, egg size can vary within a clutch [314, 823, 1218], according to geographical distribution [1067], with age [770] and race. Nonetheless, the pattern that the reserve density at birth $[E_b]$ equals the reserve density $[E]$ of the mother at egg formation generally holds; this not only removes a parameter, but also has the nice implication that the von Bertalanffy growth curve applies from birth on, at constant food density. Embryos don't feed, $f = 0$, and even embryos of endothermic species don't allocate to heating, $L_T = 0$.

The state variables (E_H, E, L) evolve from $(0, E_0, 0)$ at age $a = 0$ to $(E_H^b, [E_b]L_b^3, L_b)$ at age $a = a_b$, the age at birth. To find E_0, a_b, and L_b, given E_H^b and $[E_b]$, is a bit of a challenge, which has been overcome only recently [645].

For this purpose, it is most convenient to remove parameters by scaling to dimensionless quantities: (τ, u_E, l, u_H), see Table 2.1. The reformulated problem is now: find τ_b, l_b, u_E^0 given u_H^b, k, g, κ and $u_E^b = e_b l_b^3/g$, where e_b is the scaled reserve density at birth.

For the variable (τ, u_E, l, u_H) evolving from the value $(0, u_E^0, 0, 0)$ to the value $(\tau_b, u_E^b, l_b, u_H^b)$, the scaled model amounts to

$$\frac{d}{d\tau}u_E = -u_E l^2 \frac{g+l}{u_E + l^3} \tag{2.26}$$

$$\frac{d}{d\tau}l = \frac{1}{3}\frac{g u_E - l^4}{u_E + l^3} \tag{2.27}$$

$$\frac{d}{d\tau}u_H = (1-\kappa)u_E l^2 \frac{g+l}{u_E + l^3} - k u_H \tag{2.28}$$

or alternatively for variable (τ, e, l, e_H) evolving from the value $(0, \infty, 0, e_H^0)$ to the value $(\tau_b, e_b, l_b, e_H^b)$

$$\frac{d}{d\tau}e = -g\frac{e}{l} \tag{2.29}$$

$$\frac{d}{d\tau}l = \frac{g}{3}\frac{e-l}{e+g} \tag{2.30}$$

$$\frac{d}{d\tau}e_H = (1-\kappa)\frac{ge}{l}\frac{l+g}{e+g} - e_H\left(k + \frac{g}{l}\frac{e-l}{e+g}\right) \tag{2.31}$$

where the initial scaled maturity density $e_H^0 = (1-\kappa)g$ is such that $\frac{d}{d\tau}e_H(0) = 0$, otherwise $\frac{d}{d\tau}e_H(0) = \pm\infty$.

If $k = 1$, so $\dot{k}_J = \dot{k}_M$, we have $e_H(\tau) = e_H^0$ for all τ and $u_H(\tau) = (1-\kappa)l_b^3$. In other words: the maturity density remains constant, so maturity exceeds threshold values when structure exceeds threshold values. We then have the relationship for the structural volume at birth

$$V_b = L_b^3 = \frac{E_H^b/[E_m]}{(1-\kappa)g} \tag{2.32}$$

For $k > 1$, e_H is decreasing in (scaled) age, and for $k < 1$ increasing.

Figure 2.12 shows that data on embryo weight, yolk and respiration are in close agreement with model expectations. As is discussed later, {147}, respiration is taken to be proportional to the mobilisation rate. The two or three curves per species have been fitted simultaneously by Zonneveld [1296], and the total number of parameters is five excluding, or seven including, respiration. This is fewer than three parameters per curve and thus approaches a straight line for simplicity when measured this way. I have not found comparable data for plant seeds, but I expect a very similar pattern of development.

The examples are representative of the data collected in Table 2.2, which gives parameter estimates of some 40 species of snails, fish, amphibians, reptiles and birds. The model tends to underestimate embryo weight and respiration rate in the early phases of development. This is partly because of deviations in isomorphism, the contributions of extra-embryonic membranes (both in weight and in the mobilisation of energy reserve) and the loss of water content during development. The parameter estimates

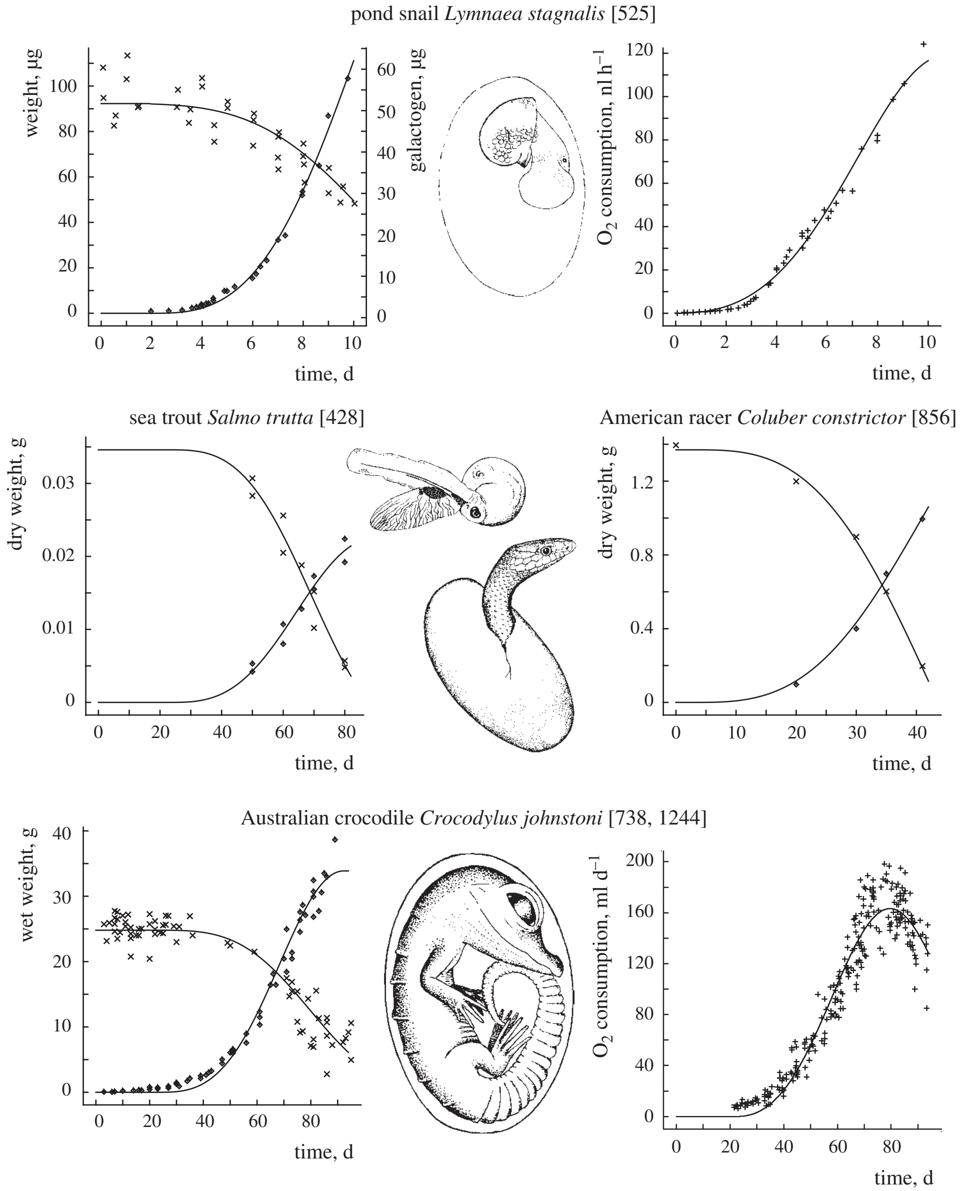

Figure 2.12 Yolk-free embryo weight (⋄), yolk weight (×) and respiration rate (+) during embryo development, and fits on the basis of the DEB model. Data sources are indicated.

for altricial birds such as the parrot *Agapornis* should be treated with some reservation, because neglected acceleration caused by the temperature increase during development substantially affects the estimates, as discussed on {173}.

The values for the energy conductance $\dot{v}$, as given in Table 2.2, are in accordance with the average value for post-embryonic development, as given on {312}, which indicates

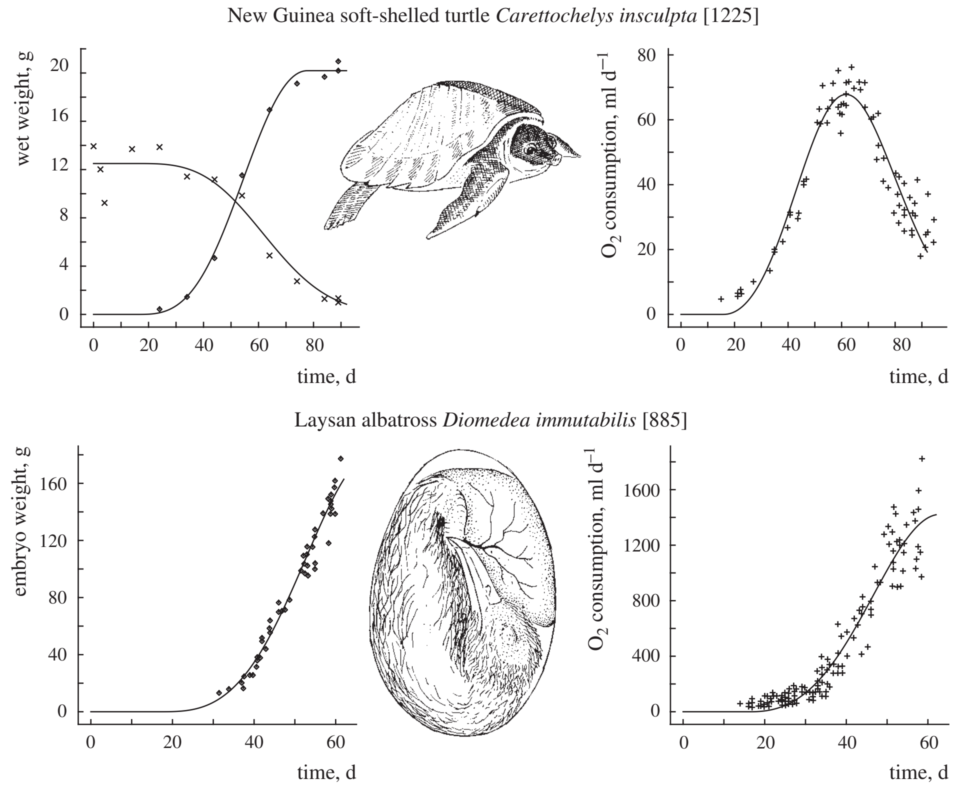

Figure 2.12 (cont.)

that no major changes in energy parameters occur at birth. The maintenance rate constant $\dot{k}_M$ for reptiles and birds is about 0.08 d^{-1} at 30 °C, implying that the energy required to maintain tissue for 12 days at 30 °C is about equal to the energy necessary to synthesise the tissue from the reserve. The maintenance rate constant for freshwater species seems to be much higher, ranging from 0.3 to 2.3 d^{-1}. Data from Smith [1081] on the rainbow trout *Oncorhynchus mykiss* result in 1.8 d^{-1} and Figure 2.10 gives 4.06 d^{-1} at 20 °C, which corresponds to some 8.5 d^{-1} at 30 °C (if it would survive that) for the waterflea *Daphnia magna*. The costs of osmosis might contribute to these high maintenance costs, as has been suggested on {45}, but for Ecdysozoa (to which *Daphnia* belongs), moulting might be costsly. The high value for *Oikopleura*, see Figure 2.19, probably relates to house production. Although information on parameter values is still sparse, it indicates that no (drastic) changes in these values occur at the transition from the embryonic to the juvenile state.

The general pattern of embryo development in eggs is characterised by unrestricted fast development during the first part of the incubation period (once the process has started) due to unlimited energy supply, at a rate that would be impossible to reach if the animal had to refill reserves by feeding. This period is followed by a retardation of

Table 2.2 Survey of re-analysed egg data, and parameter values standardised to a temperature of 30 °C, taken from [1296].

species	temp. °C	type of data	$\dot{v}_{30}$ mm d^{-1}	$\dot{k}_{M\,30}$ d^{-1}	E_b/E_0	reference
Lymnaea stagnalis	23	ED, galac, O	0.80	2.3	0.55	[527]
Salmo trutta	10	ED, YD	3.0	0.31	0.37	[430]
Rana pipiens	20	EW, O	2.5		0.87	[40]
Crocodylus johnstoni	30	EW, YW	1.9	0.060	0.31	[740]
	29, 31	O				[1245]
Crocodylus porosus	30	EW, YW	2.7	0.024	0.19	[1227]
	30	O				*1*
Alligator mississippiensis	30	EW, YW	2.7		0.34	[256]
	30	O				[1153]
Chelydra serpentina	29	ED, YD	1.9		0.35	[859]
	29	O				[399]
Carettochelys insculpta	30	EW, YW, O	1.9	0.040	0.08	[1226]
Emydura macquarii	30	EW, O	1.6	0.14	0.35	[1153]
Caretta caretta	28–30	EW, O	3.0		0.65	[4, 3]
Chelonia mydas	28–30	EW, O	3.0		0.57	[4, 3]
Amphibolurus barbatus	29	ED, YD	0.92	0.061	0.47	[860]
Coluber constrictor	29	ED, YD	1.4		0.69	[861]
Sphenodon punctatus	20	HW, O	0.85	0.062	0.25	*2*
Gallus domesticus	39	EW, O, C	3.2	0.039	0.34	[987]
	38	EW, C	3.4		0.52	[125]
Leipoa ocellata	34	EE, YE, O	1.7	0.031	0.55	[1200]
Pelicanus occidentalis	36.5	EW, O	3.2	0.10	0.77	[61]
Anous stolidus	35	EW, O	2.0	0.11	0.59	[887]
Anous tenuirostris	35	EW, O	1.8	0.20	0.59	[887]
Diomedea immutabilis	35	EW, O	2.5	0.069	0.57	[886]
Diomedea nigripes	35	EW, O	2.5	0.049	0.58	[886]
Puffinus pacificus	38	EW, O	0.92	0.084	0.61	[5]
Pterodroma hypoleuca	34	EW, O	1.9		0.20	[886]
Larus argentatus	38	EW, C	2.7	0.15	0.56	[293]
Gygis alba	35	EW, O	1.4		0.53	[885]
Anas platyrhynchos	37.5	EW	2.5	0.10	0.67	[921]
	37.5	O				[578]
Anser anser	37.5	EW	4.1	0.039	0.23	[986]
	37.5	O				[1198]
Coturnix coturnix	37.5	EW, O	1.7		0.49	[1198]
Agapornis personata	36	EW, O	0.8		0.79	[170]
Agapornis roseicollis	36	EW, O	0.84		0.81	[170]

Table 2.2 (cont.)

species	temp. °C	type of data	$\dot{v}_{30}$ mm d^{-1}	$\dot{k}_{M\,30}$ d^{-1}	E_b/E_0	reference
Troglodytes aëdon	38	EW, O	1.4		0.82	[590]
Columba livia	38	EW	2.7		0.80	[582]
	37.5	O				[1198]

EW: Embryo Wet weight
EE: Embryo Energy content
O: Dioxygen consumption rate
YW: Yolk Wet weight
YE: Yolk Energy content
C: Carbon dioxide prod. rate
ED: Embryo Dry weight
YD: Yolk Dry weight
HW: Hatchling Wet weight
'galac': galactogen content
1 P. J. Whitehead, pers. comm., 1989 ; *2* M. B. Thompson, pers. comm., 1989.

development due to the increasing depletion of energy reserves. Apart from the reserves of the juvenile, the model works out very similar to that of Beer and Anderson [74] for salmonid embryos.

In view of the goodness of fit of the model in species that do not possess shells (see the turtle data), retardation is unlikely to be due to limitation of gas diffusion across the shell, as has been frequently suggested for birds [932]. The altricial and precocial modes of development have been classified as being basically different; the precocials show a plateau in respiration rates towards the end of the incubation period, whereas the altricials do not. Figure 2.13 shows that this difference can be traced back to the simple fact that altricial birds hatch relatively early. A frequently used argument for diffusion limitation is the strong negative correlation between diffusion rates across the egg shell and diffusion resistance, when different egg sizes are compared, ranging from hummingbirds to ostriches; the product of diffusion rate and resistance does not vary a lot. This correlation probably results from a minimisation of water loss by eggs. Data from fossil plants from Greenland show a dramatic drop in stomata frequency at the end of the Triassic (208 Ma ago), which has been linked to a steep rise of the atmospheric carbon dioxide concentration to three times the present values, possibly as a result of the activity of volcanoes that mark the break-up of Pangaea [75, p 98]. The plants no longer needed many stomata, and reduced the number to reduce the water loss by evaporation.

To find τ_b, l_b, u_E^0, I first observe that from Table 2.1 and ODEs (2.26–2.27), we have

$$\frac{d}{d\tau}x = gx\frac{1-x}{l}; \quad \frac{d}{d\tau}l = \frac{g - xg - lx}{3}; \quad \frac{d}{d\tau}\alpha = \frac{x^{1/3}}{1-x}\frac{d}{d\tau}x \tag{2.33}$$

so

$$\alpha = 3g(u_E^0)^{-1/3} + B_x\left(\frac{4}{3}, 0\right) \tag{2.34}$$

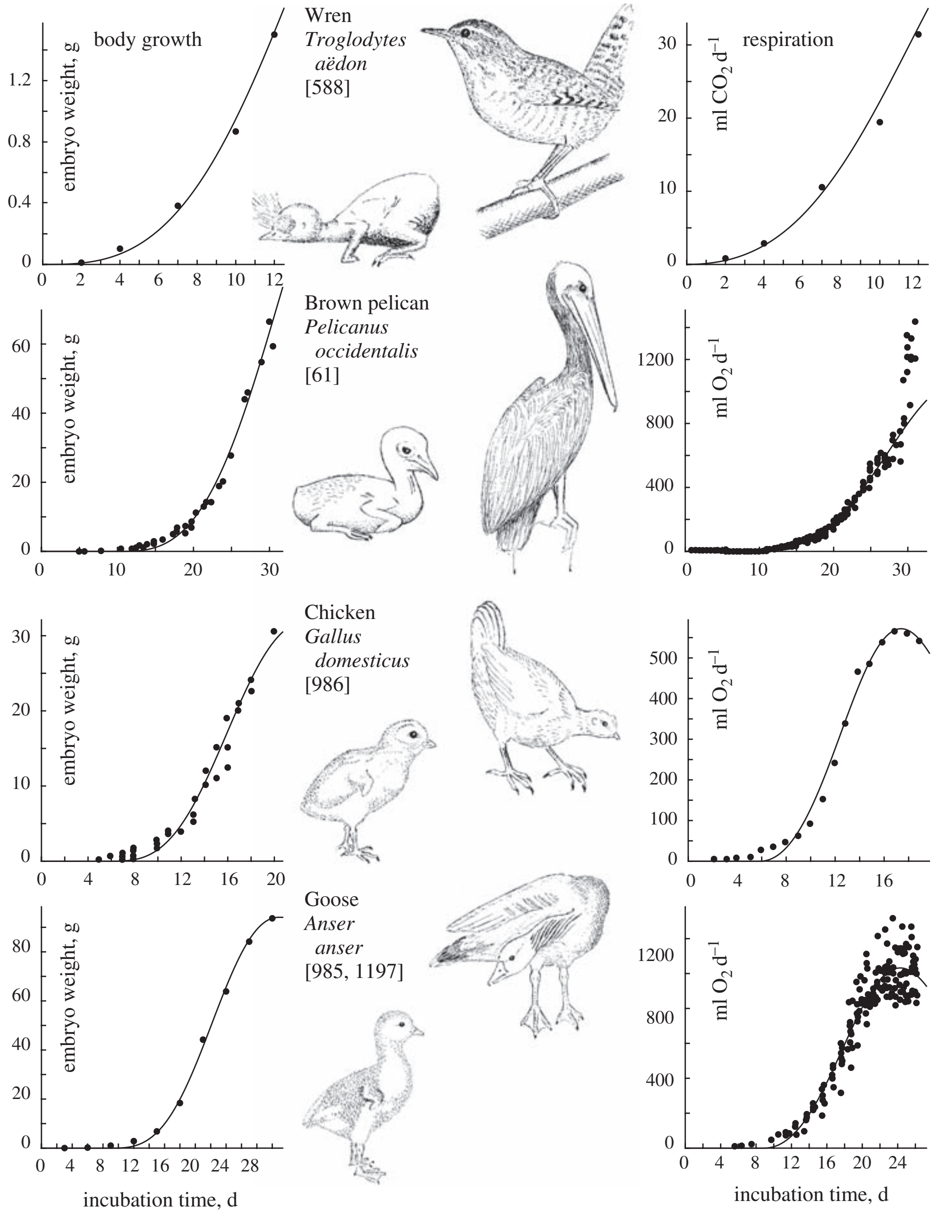

Figure 2.13 The embryonic development of altricial (wren, pelican) and precocial (chicken, goose) birds. Data from the sources indicated; fits are on the basis of the DEB model (parameters in Table 2.2). The underestimation of the initial development possibly relates to embryonic membranes. Pelican's high respiration rate just prior to hatching is attributed to internal pipping, which is not modelled. The drawings show hatchlings and adults.

for the incomplete beta function

$$B_x(\frac{4}{3},0) \equiv \int_0^x y^{1/3}(1-y)^{-1}\,dy \tag{2.35}$$
$$= \sqrt{3}\left(\arctan\frac{1+2x^{1/3}}{\sqrt{3}} - \arctan\frac{1}{\sqrt{3}}\right) + \frac{1}{2}\log(1+x^{1/3}+x^{2/3}) - \log(1-x^{1/3}) - 3x^{1/3}$$

Consequently we have

$$\alpha_b - \alpha = B_{x_b}(\frac{4}{3},0) - B_x(\frac{4}{3},0) \tag{2.36}$$
$$\frac{1}{l} = \frac{1}{l_b}\left(\frac{x_b}{x}\right)^{1/3} - \frac{B_{x_b}(\frac{4}{3},0) - B_x(\frac{4}{3},0)}{3gx^{1/3}} \tag{2.37}$$

We need this expression for $l(x)$ later in the derivation of l_b.

Scaled age at birth τ_b

The scaled age at birth τ_b follows from (2.33) and (2.37) by separation of variables and integration

$$\tau_b = 3\int_0^{x_b} \frac{dx}{(1-x)x^{2/3}(\alpha_b - B_{x_b}(\frac{4}{3},0) + B_x(\frac{4}{3},0))} \tag{2.38}$$

Notice that τ_b requires l_b in α_b, which is given below.

The parameter values in Figure 2.10 for *D. magna* imply an incubation time of 0.76 d at $f = 1$ to 0.80 d at $f = 0.5$. The eggs are deposited in the brood pouch just after moulting and develop there till birth just before the next moult, some 1.5 to 2 d later at 20 °C. This suggests a food-level-dependent diapause of around 0.5 d in *D. magna*.

The age at birth simplifies for small g and large $\dot{k}_M$, while $\dot{r}_B = \frac{\dot{k}_M g}{3(e_b+g)}$ remains fixed [661]:

$$a_b = \frac{3}{\dot{k}_M}\int_0^{x_b} \frac{dx}{(1-x)x^{2/3}(3gx_b^{1/3}l_b^{-1} - B_{x_b}(\frac{4}{3},0) + B_x(\frac{4}{3},0))} \tag{2.39}$$
$$\overset{g,\dot{k}_M^{-1}\text{ small}}{\simeq} \frac{1}{3e_b\dot{r}_B}\int_0^{x_b} \frac{dx}{(1-x)x^{2/3}x_b^{1/3}(l_b^{-1} - (x_b^{4/3} - x^{4/3})/(4e_b))}$$
$$\overset{g,\dot{k}_M^{-1}\text{ very small}}{\simeq} \frac{l_b}{3e_b\dot{r}_B}\int_0^{x_b} \frac{dx}{(1-x)x^{2/3}x_b^{1/3}} \overset{g,\dot{k}_M^{-1}\to 0}{=} \frac{l_b}{e_b\dot{r}_B} \overset{e_b=f}{=} \frac{3l_b}{\dot{k}_M g}\left(1+\frac{g}{f}\right) \tag{2.40}$$

where $x_b \equiv \frac{g}{e_b+g}$. The significance of this result is in the fact that for fixed $\dot{r}_B$, L_b and L_∞, $g \to 0$ while $\dot{k}_M \to \infty$ if a_b is running from 0 to this upper boundary. See Figure 2.14 for a graphical interpretation.

The chameleon *Furcifer labordi* is reported to live 8–9 months as egg and only 4–5 months as juvenile plus adult [577]. The males grow from some 3 cm to 10.2 cm (snout-to-vent length) with a von Bertalanffy growth rate of 0.035 d^{-1}, and the females from

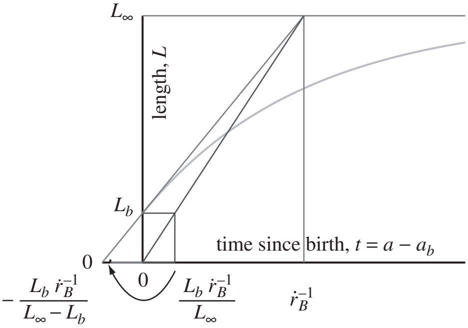

Figure 2.14 The von Bertalanffy growth curve $\frac{d}{dt}L = \dot{r}_B(L_\infty - L)$, with the graphical interpretation of the von Bertalanffy growth rate $\dot{r}_B$, and the maximum possible age at birth $\frac{L_b \dot{r}_B^{-1}}{L_\infty}$ in the context of DEB theory. The tangent line at $t = 0$ intersects the asymptote at time $\dot{r}_B^{-1}$; the line from the origin to this intersection point hits level L_b at time $\frac{L_b}{L_\infty \dot{r}_B}$, which is the maximum possible incubation time.

2.6 cm to 7.8 cm with a von Bertalanffy growth rate of 0.04 d^{-1}. The maximum age at birth that is consistent with DEB theory is thus some 8.3 d, rather than the observed 8–9 months. The eggs are buried in the soil, where it might be some 10 °C cooler than during post-embryonic growth. Even after correction for this difference, this suggests that this species is on metabolic hold for most of the time.

The foetal special case, where $a_b = 3L_b/\dot{v} = \frac{3l_b}{\dot{k}_M g}$, represents a lower boundary for the age at birth (of eggs). So the possible range for a_b is

$$\frac{3l_b}{\dot{k}_M g} < a_b < \frac{3l_b}{\dot{k}_M g}(1 + g/e_b) \quad \text{or} \quad 1 < a_b \frac{\dot{k}_M g}{3l_b} < 1 + g/e_b \tag{2.41}$$

Scaled initial amount of reserve u_E^0

The scaled initial amount of reserve u_E^0 directly follows from (2.34) for $x = x_b$ and $\alpha = \alpha_b$

$$u_E^0 = \left(\frac{3g}{\alpha_b - B_{x_b}\left(\frac{4}{3}, 0\right)} \right)^3 \quad \text{so} \quad E_0 = u_E^0 g [E_m] L_m^3 \tag{2.42}$$

which again requires l_b in α_b.

The parameter values in Figure 2.10 for *D. magna* imply a scaled initial amount of reserve of $U_E^0 = 0.04\,\mathrm{mm^2 d}$ at $f = 1$ and $0.0246\,\mathrm{mm^2 d}$ at $f = 0.5$ at 20 °C. The scaled amount at birth equals $U_E^b = f[E_m]L_b^3 = 0.0313$ and $0.0156\,\mathrm{mm^2 d}$, respectively. So a fraction of 0.79 and 0.63 for $f = 1$ and 0.5, respectively, of the initial reserve is still left at birth. Figure 2.10 has, however, no data on embryo development or reserve, which demonstrates the strength of DEB theory.

Data on embryo development, Table 2.2, also show that about half of the reserves are used during embryonic development. The deviating values for altricial birds are artifacts, caused by the above-mentioned acceleration of development by increasing temperatures. Congdon *et al.* [217] observed that the turtles *Chrysemus picta* and *Emydoidea blandingi* have 0.38 of the initial reserves at birth. Respiration measurements on seabirds by Pettit *et al.* [888] indicate values that are somewhat above the ones reported in the table. The extremely small value for the soft-shelled turtle, see also

Figure 2.12, relates to the fact that these turtles wait for the right conditions to hatch, after which they have to run the gauntlet as a cohort at night from the beach to the water, where a variety of predators wait for them.

Scaled length at birth l_b

For the variable (τ, e_H) evolving from the value $(0, e_H^0)$ to the value (τ_b, e_H^b) we have

$$\frac{d}{d\tau} e_H = (1-\kappa) g (1-x) \left(\frac{g}{l(x)} + 1 \right) - e_H \left(k - x + g \frac{1-x}{l(x)} \right) \tag{2.43}$$

Now consider the variable (x, e_H) evolving from the value $(0, e_H^0)$ to the value (x_b, e_H^b) or the variable (x, y) evolving from the value $(0, 0)$ to the value (x_b, y_b):

$$\frac{d}{dx} e_H = \frac{e_H^0}{x} \left(\frac{l(x)}{g} + 1 \right) - \frac{e_H}{x} \left(\frac{k-x}{1-x} \frac{l(x)}{g} + 1 \right) \quad \text{for} \quad e_H^0 = e_H(0) = (1-\kappa) g$$

$$\frac{d}{dx} y = r(x) - y s(x) \quad \text{for} \quad r(x) = g + l(x); \quad s(x) = \frac{k-x}{1-x} \frac{l(x)}{gx} \tag{2.44}$$

where $l(x)$ is given in (2.37). The ODE for y can be solved to

$$y(x) = v(x) \int_0^x \frac{r(x_1)}{v(x_1)} dx_1 \quad \text{with} \quad v(x) = \exp(- \int_0^x s(x_1)\, dx_1) \tag{2.45}$$

The quantity l_b must be solved from $y_b = y(x_b) = g x_b v_H^b l_b^{-3}$, see Table 2.1. So we need to find the root of t as function of l_b with

$$t(l_b) = \frac{x_b g v_H^b}{v(x_b) l_b^3} - \int_0^{x_b} \frac{r(x)}{v(x)} dx = 0 \tag{2.46}$$

From this equation it becomes clear that the parameters κ and u_H^b affect l_b only via $v_H^b = \frac{u_H^b}{1-\kappa}$; a conclusion that is more difficult to obtain using the ODE for the scaled maturity density e_H rather than that for abstract variable y. Notice that the solution of l_b (and that of u_E^0 and τ_b) for the boundary value problem for the ODE for (u_E, l, e_H) as given in (2.26–2.28) depends on the four parameters g, k, v_H^b and e_b only. The solution for l_b must be substituted into (2.42) to obtain u_E^0 and in (2.38) to obtain τ_b; the scaled reserve at birth is $u_E^b = e_b l_b^3 / g$.

The parameter values in Figure 2.10 for *D. magna* imply a birth length of 0.686 and 0.685 mm at $f = 1$ and 0.5, respectively. These values hardly differ much because $\dot{k}_J$ is close to $\dot{k}_M$.

Special case $e \to \infty$: foetal development

Foetal development differs from that in eggs in that energy reserves are supplied continuously via the placenta. The feeding and digestion processes are not involved. Otherwise, foetal development is taken to be identical to egg development, with initial reserves that can be taken to be infinitely large, for practical purposes. At birth, the neonate receives an amount of reserves from the mother, such that the reserve density of the neonate equals that of the mother. So the approximation $[E] \to \infty$ or $e \to \infty$ for the foetus can be made for the whole gestation period, because the foetus lives on the reserves of the mother. In other words: unlike eggs, the development of fetuses is not restricted by energy reserves. Initially the egg and foetus develop in the same way, but the foetus keeps developing at a rate not restricted by the amount of reserves till the end of the gestation time, while the development of the egg becomes retarded, due to depletion of the reserves.

The special case $e \to \infty$ means that $\frac{d}{d\tau} l = g/3$, or $l(\tau) = g\tau/3$. The foetal structural volume thus behaves as

$$\frac{d}{dt} V = \dot{v} V^{2/3} \quad \text{so} \quad V(t) = (\dot{v}t/3)^3 \tag{2.47}$$

This growth curve was proposed by Huggett and Widdas [536] in 1951. Payne and Wheeler [874] explained it by assuming that the growth rate is determined by the rate at which nutrients are supplied to the foetus across a surface that remains in proportion to the total surface area of the foetus itself. This is consistent with the DEB model, which gives the energy interpretation of the single parameter. The graph of foetal weight against age resembles an exponential growth curve, but in fact it is less steep; the model has the property that subsequent weight doubling times increase by a factor $2^{1/3} = 1.26$, while there is no increase in the case of exponential growth.

The fit is again excellent; see Figure 2.15. It is representative for the data collected in Table 2.3 taken from [1296]. A time lag for the start of foetal growth has to be incorporated, and this diapause may be related to the development of the placenta, which possibly depends on body volume as well. The long diapause for the grey seal *Halichoerus* probably relates to timing with the seasons to ensure adequate food supply for the developing juvenile. Variations in weight at birth are primarily due to variations in gestation period, not in foetal growth rate. For comparative purposes, energy conductance $\dot{v}$ is converted to 30 °C, on the assumption that the Arrhenius

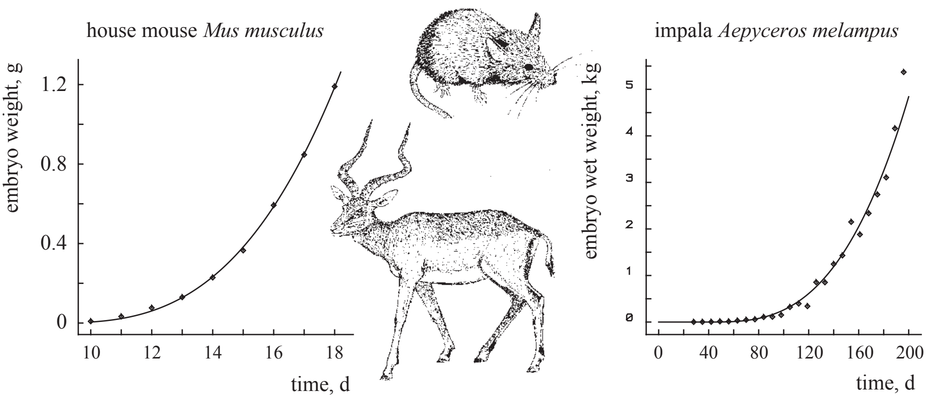

Figure 2.15 Foetal weight development in mammals, see Figure 6.2. Parameters are given in Table 2.3.

Table 2.3 The estimated diapause t_0, age at birth a_b, scaled energy conductance $\dot{v}^* = \dot{v}(1+\omega_w)^{1/3}$, birth weight W_b for mammals; specific density $d_V = 1\,\mathrm{g\,cm^{-3}}$, $e = 1$.

species	t_0 d	a_b d	$\dot{v}^*$ $\mathrm{cm\,d^{-1}}$	W_b g	reference
Homo sapiens	26.8	259	0.180	3750	[1230]
Oryctolagus cuniculus	10.7	19.2	0.560	46.2	[695]
Lepus americanus	13.1	21.1	0.573	66	[127]
Cavia porcellus	15.7	48.8	0.269	84	[291]
	19.5	48.5	0.239	101	[541]
Cricetus auratus	9.29	5.8	0.570	1.43	[925]
Mus musculus	8.2	11.8	0.278	1.2	[728]
Rattus norvegicus					
Wistar	11.4	7.7	0.487	2	[346]
	11.8	9.3	0.525	4.3	[536]
albino	12.4	9.5	0.542	5	[355]
Clethrionomys glareolus	8.29	10.4	0.374	2.2	[224]
Aepyceros melampus	39.4	166	0.316	5370	[331]
Odocoileus virginianus	34.9	155	0.296	3577	[977]
Dama dama	9.94	155	0.345	56400	[35]
Cervus canadensis	24.9	217	0.336	14500	[811]
Lama pacus	7.47	90.7	0.120	47.7	[347]
Ovis aries					
Welsh	43.9	105.8	0.482	4913	[536]
	33.3	111	0.433	4120	[213]
Karakul	31.0	116	0.426	4445	[306]
	27.5	121.8	0.403	4384	[568]
Capra hircus	24.3	126.3	0.339	2910	[317]
	31.3	117.3	0.365	2910	[59]
Bos taurus	59.5	204	0.475	33750	[1261]
Equus caballus	37.0	348	0.370	79200	[789]
Sus scrofa	4.73	129	0.266	1500	[1219]
Yorkshire	5.49	121	0.283	1500	[1178]
large white	23.6	90	0.383	1500	[907]
Essex	14.1	107	0.321	1500	[907]
Felix catus	18.8	39.8	0.371	119.5	[223]
Pipistrellus pipistrellus					
1978	9.95	14.2	0.237	1.4	[927]
1979	13.7	18.5	0.181	1.4	[927]
Halichoerus grypus	145	165	0.375	8732	[501]

temperature, T_A, is 10 200 K and the body temperature is 37 °C for all mammals in the table. This is a rather crude conversion because the cat, for instance, has a body temperature of 38.6 °C. Weights were converted to volumes using a specific density of $[W_w] = 1\,\mathrm{g\,cm^{-3}}$.

One might expect that precocial development is rapid, resulting in advanced development at birth and, therefore, comes with a high value for the energy conductance. The values collected in Table 2.3, however, do not seem to have an obvious relationship with altricial–precocial rankings. The precocial guinea-pig and alpaca as well as the altricial human have relatively low values for the energy conductance. The altricial–precocial ranking seems to relate only to the relative volume at birth V_b/V_m.

We further have

$$\frac{d}{d\tau}u_H = (1-\kappa)l^2(g+l) - ku_H \tag{2.48}$$

$$u_H(\tau) = \frac{g^3(1-\kappa)}{3^3k^4}\left(k^2\tau^2(3k+k\tau-3)+6(k-1)(1-\tau-\exp(-k\tau))\right) \tag{2.49}$$

The equation $u_H(\tau_b) = u_H^b$ has to be solved numerically for τ_b, but for $k=1$ we have $u_H^b = (1-\kappa)3^{-3}g^3\tau_b^3 = (1-\kappa)l_b^3$. The solution of this equation is stable and fast; the resulting scaled length at birth $l_b = g\tau_b/3$ can be used to start the Newton Raphson procedure. This start is preferable if k is substantially different from 1. From $l_b < 1$, so $\tau_b < 3/g$, we can derive the constraint

$$\frac{k^2u_H^b}{1-\kappa} < k + g(k-1) + g^3\frac{k-1}{k^2}\,\frac{1-3/g-\exp(-3k/g)}{9/2} \tag{2.50}$$

For $u_E^b = u_E(\tau_b)$, the cost for a foetus amounts to

$$u_E^0 = u_E^b + \kappa l_b^3 + u_H^b + \int_0^{\tau_b}(\kappa l^3(\tau) + ku_H(\tau))\,d\tau = u_E^b + l_b^3 + \frac{3}{4}\frac{l_b^4}{g}; \quad E_0 = u_E^0 g[E_m]L_m^3 \tag{2.51}$$

where the five terms correspond with the costs of reserve, structure, maturity, somatic and maturity maintenance, respectively. The second equality follows from the structure of DEB theory: the investment in maturity plus maturity maintenance equals $\frac{1-\kappa}{\kappa}$ times the investment in structure plus somatic maintenance and $l(\tau) = g\tau/3$. The cost of a foetus is somewhat smaller than that of an egg (2.42).

Foetal development obviously affects the energetics of the mother. This is discussed at {282}.

2.6.3 States at puberty

Puberty occurs as soon as $E_H = E_H^p$, or $U_H = U_H^p = E_H^p/\{\dot{p}_{Am}\} = U_H^p$ or $u_H = u_H^p = \frac{E_H^p}{g[E_m]L_m^3}$. If $\dot{k}_J = \dot{k}_M$, and so maturity density is constant, and $L_T = 0$, we have

$$V_p = \frac{E_H^p/[E_m]}{(1-\kappa)g} \tag{2.52}$$

Otherwise the structural volume at puberty $V_p = L_p^3$ can vary with food density history and even more than V_b can. It must be found numerically from integration of $\frac{d}{dt}V$ to $E_H(t) = E_H^p$.

Generally little can be said about age and length at puberty, but if food density X, and so scaled functional response f, remains constant, (2.23), (2.22) and (2.43) show that age and scaled length at puberty amount to

$$a_p = a_b + \frac{1}{\dot{r}_B} \ln \frac{L_\infty - L_b}{L_\infty - L_p} \tag{2.53}$$

$$l_p = l(v_H^p) \quad \text{with} \quad \frac{d}{dv_H} l = \frac{(f - l - l_T)g/3}{fl^2(g+l) - kv_H(g+f)} \quad \text{and} \quad l(v_H^b) = l_b \tag{2.54}$$

where $k = \dot{k}_J/\dot{k}_M$ as before. This differential equation has to be integrated numerically and results in a satiating function $l_p(f)$ if $k < 1$. We must have that $f > l_p + l_T$ to allow for adolescence.

The parameter values in Figure 2.10 for *D. magna* imply a length at puberty of 2.46 and 2.45 mm at $f = 1$ and 0.5, respectively. These values hardly differ because $\dot{k}_J$ is close to $\dot{k}_M$.

2.6.4 Reduction of the initial amount of reserve

The embryonic period is elongated if the initial amount of reserve is reduced, while the cumulative energy investment to complete the embryonic stage is the same. The mechanism is that reducing the amount of reserve reduces the mobilisation of reserve, so it takes longer to reach a certain maturity level. Large eggs, so large initial energy supplies, result in short incubation times if eggs of one species are compared. Crested penguins, *Eudyptes*, are known for egg dimorphism [1218]; see Figure 2.16. They first lay a small egg and, some days later, a 1.5 times bigger one. As predicted by the DEB

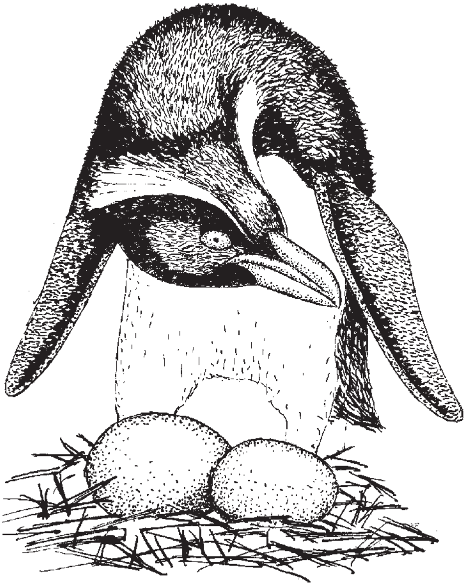

Figure 2.16 Egg dimorphism occurs as standard in crested penguins (genus *Eudyptes*). The small egg is laid first, but it hatches later than the big one, which is 1.5 times as heavy. The DEB theory explains why the large egg requires a shorter incubation period. The illustration shows the Snares crested penguin *E. atratus*.

model, the bigger one hatches first, if fertile, in which case the parents cease incubating the smaller egg, because they are only able to raise one chick. They continue to incubate the small egg only if the big one fails to hatch. This is probably an adaptation to the high frequency of unfertilised eggs or other causes of loss of eggs (aggression [1218]), which occur in this species.

Incubation periods only decrease for increasing egg size if all other parameters are constant. The incubation period is found to increase with egg size in some beetle species, lizards and marine invertebrates [210, 327, 1067]. In these cases, however, the structural biomass at hatching also increases with egg size. This is again consistent with the DEB theory, although the theory does not explain the variation in egg sizes.

Hart [466] studied the effect of separation of the embryonic cells of the sea urchin *Strongylocentrotus droebachiensis* in the two-cell stage on the energetics of larval development. Both the size and the feeding capacity of the resulting larvae were reduced by about one-half, but the time to metamorphosis is about the same (7 d at 8–13 °C). The maximum clearance rate of dwarf and normal larvae was found to be the same function of the ciliated band length. Larvae fed a smaller ration had longer larval periods, but food ration hardly affected size at metamorphosis. Egg size affected juvenile test diameter only slightly.

Armadillos typically separate cells in the four-cell stage of the embryo, giving birth to four identical offspring. Humans rarely do this successfully, then giving birth to four babies of about 1 kg each, rather than the typical 3 kg. In terms of an effect on length this reduction amounts to a factor $(1/3)^{1/3} = 0.69$. The human growth curve fits the von Bertalanffy curve very well, with a von Bertalanffy growth rate of $\dot{r}_B = \frac{\dot{k}_M g}{3(e+g)} = 0.123$ a^{-1}, see {289}. We can safely assume that the scaled reserve density was close to its maximum $e = 1$ for the post-embryonic stages. Moreover the age at birth is $a_b = \frac{3l_b}{g\dot{k}_M} = 0.75$ a for humans. If we take a typical maximum adult weight of 70 kg, then the scaled length at birth equals $l_b = (3/70)^{1/3} = 0.35$. So the energy investment ratio equals $g = \frac{l_b}{a_b \dot{r}_B} - 1 = 2.79$, the somatic maintenance rate coefficient $\dot{k}_M = \frac{3l_b}{g a_b} = 0.5$ a^{-1} and the scaled age at birth $\tau_b = a_b \dot{k}_M = 0.375$. With these values for g, e_b and l_b, the scaled cost amounts to $u_E^0 = 0.062$ from (2.51). In the case of four babies with a length reduced by a factor 0.69, the scaled cost per baby equals $u_E^0 = 0.02$, so summed over the four babies this is 1.3 times the amount of a single baby; not a surprising result, in view of the 4 kg of babies relative to the 3 kg for a single baby.

If the separation of the cells affects the required cumulative investment in development, however, other predictions result. It is then quite well possible that incubation is hardly affected, while size at birth is. Standard DEB theory correctly predicts that a reduction of the feeding level elongates the larval period and hardly affects size at puberty for a particular relationship between the somatic and maturity maintenance costs.

As discussed at {300}, the reserve density capacity $[E_m] = \{\dot{p}_{Am}\}/\dot{v}$ scales with maximum structural length of a species. So species with a larger ultimate body size tend to have a relatively larger reserve capacity. It turned out that for the combination of parameter values as found for *D. magna* we have to apply a zoom factor of at least $z = 1.87$ to arrive at a minimum maximum body size for which cell separation might be successful.

For $k > 1$, the structural volume at birth increases after halving, and decreases for $k < 1$. Since reserve contributes to weight, the weight at birth is close to half of the original weight at birth, irrespective of the value of k. The age of the two-cell stage is probably smaller than $\tau_b/3$, but the results are very similar.

The removal of an amount of reserve at the start of the development, as is frequently done [351, 555, 556, 809, 1067], elongates the incubation time (as observed in the gypsy moth [994]), and reduces the reserve at hatching. This experiment simulates the natural situation, where the nutritional status of the mother affects the initial amount of reserve. The pattern is rather similar to that of the separation of cells at an early stage, because reductions of structure and maturity at an early stage have little effect. The initial amount of reserve is a U-shaped function of the reserve at birth. The right branch is explained by the larger amount of reserve at birth, the left branch by the larger age at birth, which comes with larger cumulative somatic maintenance requirements.

The size of neonates of trout and salmon was found to increase with initial egg size [314, 478], suggesting that $k < 1$ for salmonids. This also applies to the emu *Dromaius novaehollandiae* [305], and probably represents a general pattern.

2.7 Reproduction: excretion of wrapped reserve

Organisms can achieve an increase in numbers in many ways. Sea anemones can split off foot tissue that can grow into a new individual. This is not unlike the strategy of budding yeasts. Colonial species usually have several ways of propagating. Fungi have intricate sexual reproduction patterns involving more than two sexes. Under harsh conditions some animals can switch from parthenogenetic to sexual reproduction, others develop spores or other resting phases. It would not be difficult to fill a book with descriptions of all the possibilities. The standard DEB model assumes propagation via eggs; dividing organisms don't have an embryo or adult stage, and divide when the maturity exceeds a threshold. This resets maturity and reduces the amount of structure and reserve.

Energy allocation to reproduction equals

$$\dot{p}_R = (1 - \kappa)\dot{p}_C - \dot{k}_J E_H^p \qquad (2.55)$$

cf. (2.18), where E_H is now replaced by the constant E_H^p, because the flux to maturity is redirected to reproduction at puberty, which means that maturity does not change in adults. The cost of an egg E_0 or a foetus is given in (2.42) or (2.51), so the mean reproduction rate $\dot{R}$ in terms of number of eggs per time equals